

Documentation for Deploying a Node.js Application with MySQL using systemd

Project Overview

The objective of this hands-on project is to deploy a Node.js application that interacts with a MySQL database on a Linux server. The application must be configured to start automatically at boot time using systemd and restart automatically if the application crashes.

Scenario

You are tasked with deploying a Node.js application that:

1. Connects to a MySQL database.
2. Exposes two endpoints:
 - `/health` that checks and returns the status of the database connection.
 - `/users` that retrieves and returns a list of users from the database.

Additionally, you need to configure the system such that the Node.js application starts automatically when the server boots, runs under a non-root user account, and logs output to the systemd journal.

Database Setup

1. Install MySQL:

- If MySQL is not installed, use the following commands to install it:

```
sudo apt update
sudo apt install mysql-server
```

2. Secure the MySQL Installation:

- Update the mysql ip can access only from application

3. Create the Database, User, and Table:

- Log into MySQL:

```
sudo mysql -u root -p
```

- Create the `practice_app` database:

```
CREATE DATABASE practice_app;
```

- Create the `users` table:

```
CREATE TABLE users (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  name VARCHAR(100),  
  email VARCHAR(100)  
);
```

- Insert sample data into the `users` table:

```
INSERT INTO users (name, email) VALUES  
(  
'John Doe', 'john.doe@example.com'),  
(  
'Jane Smith', 'jane.smith@example.com'),  
(  
'Alice Johnson', 'alice.johnson@example.com');
```

Application Setup

1. Create a Node.js project directory:

- Initialize the Node.js project by running:

```
mkdir nodejs-app  
cd nodejs-app  
npm init -y
```

2. Install Necessary Dependencies:

- Install `Express` and `mysql2` for connecting to the MySQL database and building the API:

```
npm install express mysql2
```

3. Create the Node.js Application:

- In the `nodejs-app` directory, create a `server.js` file with the following code:

```
const express = require("express");  
const mysql = require("mysql2"); // Ensure this is correctly imported  
  
const app = express();  
const PORT = 3000; // Change port if needed  
  
// Set the IP address manually  
const SERVER_IP = "192.168.150.131"; // Dynamically Replace with actual server IP  
  
// MySQL Configuration  
const db = mysql.createPool({  
  connectionLimit: 10,  
  host: SERVER_IP,  
  user: "app_user",  
  password: "app_user", // Set your MySQL root password here  
  database: "practice_app",  
  port: 3306  
});
```

```

// Health Check Endpoint
app.get("/health", (req, res) => {
  db.getConnection((err, connection) => {
    if (err) {
      return res.status(500).json({ status: "DOWN", error: err.message
    });
    }
    connection.release();
    res.json({ DB Status: "UP" });
  });
});

// Get Users Endpoint
app.get("/users", (req, res) => {
  db.query("SELECT * FROM users", (err, results) => {
    if (err) {
      return res.status(500).json({ error: err.message });
    }
    res.json(results);
  });
});

// Start Server
app.listen(PORT, () => {
  console.log(`Server running on http://${SERVER_IP}:${PORT}`);
});

```

Systemd Configuration

1. Create a Dedicated System User for Running the Application:

- Create a non-root user for running the application:

```
sudo useradd -r -s /bin/false nodejs
```

2. Set Permissions and Move the Application:

- Move the Node.js application to a proper directory (`/opt/app`):

```
sudo mkdir -p /opt/app
sudo mv /path/to/your/nodejs-app /opt/app/
sudo chown -R nodejs:nodejs /opt/app
```

3. Create a systemd Service File:

- Create a `nodejs-app.service` file in `/etc/systemd/system/` with the following contents:

```
[Unit]
Description=Node.js Application
After=network.target

[Service]
ExecStart=/usr/bin/node /opt/app/server.js
Restart=always
User=nodejs
Group=nodejs
WorkingDirectory=/opt/app
StandardOutput=syslog
StandardError=syslog
SyslogIdentifier=nodejs-app

[Install]
WantedBy=multi-user.target
```

4. Enable and Start the Service:

- Enable the service to start at boot time and start it manually:

```
sudo systemctl enable nodejs-app  
sudo systemctl start nodejs-app
```

5. Verify the Service:

- Check the status of the service to ensure it is running:

```
sudo systemctl status nodejs-app
```

Conclusion

This project covers the deployment of a Node.js application with MySQL on a Linux server using systemd. By following the steps outlined above, the application can be deployed to automatically start at boot time, restart if it crashes, and run securely under a non-root user. The systemd service ensures the application is always available, even in case of failure.